AutoMD-SAXS Manual

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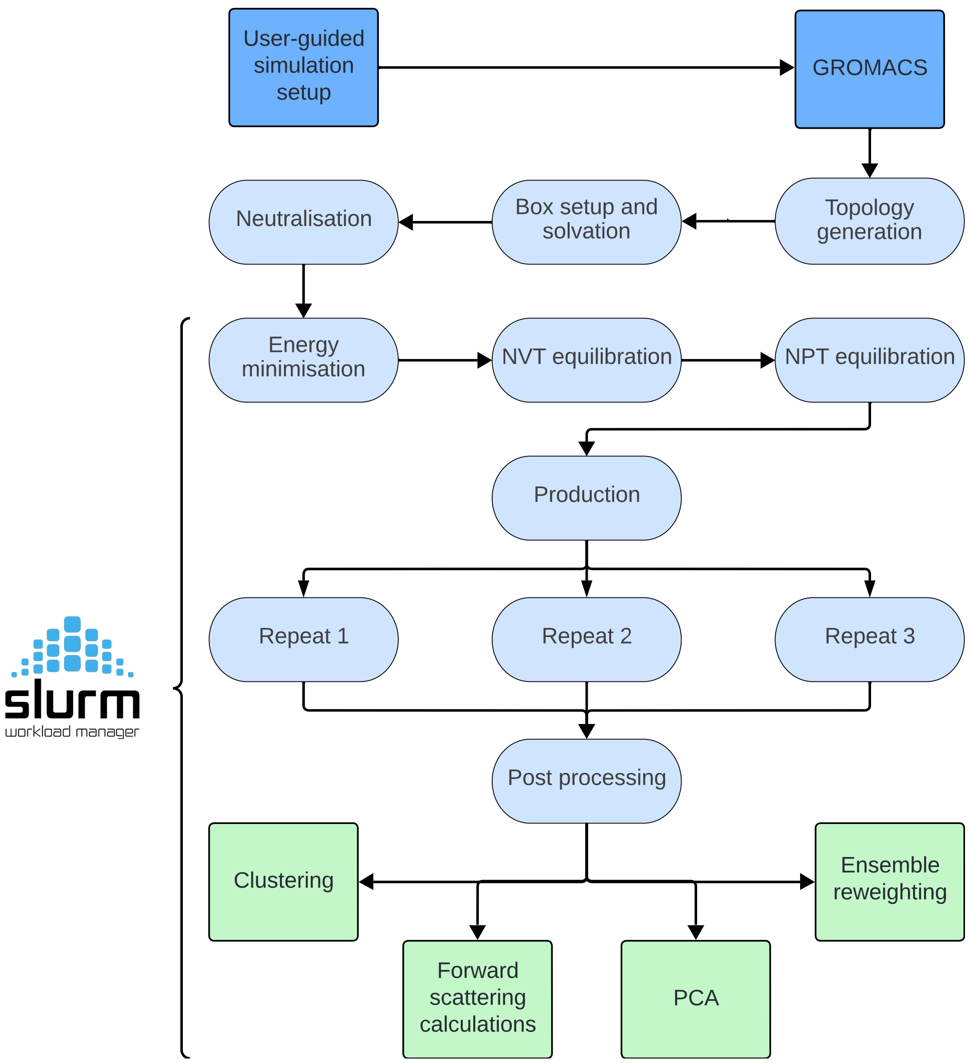

Protein preparation - an essential first step

Pre-processing protein structures is a common challenge faced during the setup of MD simulations. Errors in the starting file may result in erroneous simulation results that don't represent the system of interest. Various inconsistencies can arise when using a structure generated by X-ray crystallography for example, rarely resolving hydrogen atoms due to their poor diffraction power, as well as often failing to resolve protein loops. As a result, prior to simulation, structure files should undergo rigorous examination and preparation to ensure they correspond to the sequence of interest and take on a physically realistic conformation.

If integrating SAXS data into your simulation analysis, it is imperative that the sequence of the protein model is identical to sequence used in the SAXS experiment. For instance, if a HIS-tag was used in the SAXS experiment, but is missing in the protein model, it should be modelled in. This software package does not provide an automated functionality for modelling of missing residues.

AutoMD-SAXS was initially designed for implementation at Beamline 21, Diamond Light Source. At Diamond, Protein Preparation Wizard (PPW) is used for automated preparation. Users without access to PPW who wish to use AutoMD-SAXS are advised to approach the protein preparation steps one of two ways. If intending to simulate using the charmm36m force field, prepare your system using charmm gui (https://www.charmm-gui.org/). If intending to use the amber14sb force field, protonate your structure using the H++ webserver (http://newbiophysics.cs.vt.edu/H++/). Validity of PDB models should be vigorously checked prior to MD, with particular care given to the protonation states of titratable residues, and rotamertic states. Once you are content with your structure, it can be copied (along with optional SAXS data) to the `AutoMD-SAXS/` directory.

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Two scripts are used to run AutoMD-SAXS: `simulation_setup.sh` and `run_MD.sh`. Both are outlined here.

`simulation_setup.sh`

The `simulation_setup.sh` script is the entry point for preparing your system's configurations before launching `run_MD.sh`. It guides the user through a series of interactive prompts, captures each choice, and writes a unified `configurations.txt` file that all downstream scripts will source. Upon running `simulation_setup.sh`, the user is given a list of prompts:

1. GROMACS Module Management

- **Prompt:** "Do you load GROMACS via `module load`? (y/N)"
- **Behaviour:**
  - If `Y`, the user specifies the exact module name (e.g. `gromacs/2021.2/intel`), and the script generates helper functions `load_gmx()` / `unload_gmx()`.
  - If `N` (default), the helpers become no-ops, relying on a locally installed `gmx_mpi` in the `$PATH`.
- **Outcome:** A wrapper stored in `conifgurations.txt` for invoking GROMACS.

2. Python Environment Selection

- **Prompt:** "Full path to the Python interpreter for the automdsaxs env (e.g. '/home/you/.conda/envs/automdsaxs/bin/python')?"
- **Behaviour:**
  - If a path is provided, the script records it as `PYTHON_CMD`
- **Outcome:** Creates direct link to required environment. Eliminates "module-mix" errors by guaranteeing that every Python invocation uses the intended interpreter and package set.

3. System Type & Force-Field Selection

Prompt: "Select your system type: 1) Protein 2) Protein–ligand"

Behaviour:
- Selecting 1) Protein, the script prompts: "Select your force field: 1) `amber14sb` 2) `charmm36m`" Choice sets the `FORCE_FIELD` variable accordingly.
- Selecting 2) Protein–ligand, no further prompt is needed. `FORCE_FIELD` is fixed to `amber14sb`, and GAFF2 (via ACPYPE) will parameterise the ligand. Current version of AutoMD-SAXS does not support protein-ligand systems with CHARMM force fields.

Outcome:
- `SYSTEM` is exported as either "Protein" or "Protein–ligand."
- `FORCE_FIELD` is exported as "amber14sb" or "charmm36m".

4. Protein Preparation Wizard

Prompt: "Did you use the Protein Preparation Wizard to prepare your structure for MD? (y/N)"

Behaviour:
- If `Y`, the script assumes standard PPW naming conventions and applies the corresponding renaming logic.

- If `N`, alternative mapping rules are applied (e.g. for H++ or CHARMM-GUI outputs) to ensure all atoms and residues still match the selected force field.

Outcome:

- Exports `PPW=yes` or `PPW=no`.
- Selects the correct conversion directory (`$FF_CONVERT/amber` vs. `$FF_CONVERT/non_ppw_convert`).
- Guarantees that downstream topology generation and MD input files use the proper atom/residue naming scheme for the chosen preparation method.

5. Box Geometry

Prompt: "Choose the shape of the periodic simulation box: 1) Dodecahedron (globular particles) 2) Rectangular (anisotropic rod-shaped particles)"

Behaviour:

- If `1` is selected, `BOX_SHAPE="dodecahedron"`
- If `2` is selected, `BOX_SHAPE="rectangular"`

Outcome:

- Exports `BOX_SHAPE` for use in all `editconf` and subsequent solvation steps.
- Ensures an appropriate solvent buffer and minimises wasted box volume.

6. Ionic Strength (mM)

Prompt: "Choose your ionic concentration (mM) e.g. 0.15"

Behaviour:

- For SAXS-integrated simulations, enter the actual experimental buffer ionic strength.
- For standard MD without SAXS, a value of 150 mM (100–200 mM range) ensures human physiological conditions.

Outcome:

- Exports `IONIC_CONCENTRATION` variable
- The pipeline uses this value when calling `genion` to replace water molecules with Na^+/Cl^- ions and achieve the specified ionic strength.

7. Select simulation length

Prompt: "Select simulation length (nanoseconds)"

Behaviour:

- Creates `SIMULATION_TIME` variable and uses it to calculate `nsteps` in production MDP file.

Outcome:

- Three production replicates are submitted downstream for the specified length.

8. Maximum Scattering Dimension (D_{max})

Prompt: "What is the maximum scattering dimension (D_{max}) described by the experimental SAXS data?"

Behaviour:

- If integrating SAXS data, the user supplies the experimentally determined $\mathrm{D}_{\mathrm{max}}$.
- If running without SAXS, this prompt is not invoked and a 2nm buffer between protein and box edge is used.
- The pipeline calculates the required box buffer as:

$$\text{buffer} = 2\text{ nm} + |\mathrm{D}_{\mathrm{max,exp}} - \mathrm{D}_{\mathrm{max,model}}|.$$

Outcome:

- Exports `DMAX` for use in box-sizing
- Ensures the simulation box padding is large enough to account fo differences in model size experimental D_{max}

9. Disulfide Bond Definitions

Prompt: "Does your system contain disulfide bonds? (y/N)"

Behaviour:

- If `y`, `DISULFIDE=yes`, if `n`, `DISULFIDE=no`

Outcome:

- If `DISULFIDE=yes`, applies `-ss` tag to `pdb2gmx` during topology generation. If `DISULFIDE=no`, does not apply `-ss` flag

10. Email Notifications

Prompt: "Would you like to receive an email upon simulation completion? (y/N)"

Behaviour:
- If y, you are prompted to enter your email address.
- Exports EMAIL_ADDR="<your_address>" for use in all SBATCH submissions
- Generates SLURM flags –mail-type=ALL and –mail-user=<your_address>.

Outcome:
- Email notifications are sent at initiation and completion of post_processing.sh (final step).

11. SLURM Partition Selection

Prompt: "Would you like to specify the SLURM partition now? (y/N)"

Behaviour:
- If y, you are prompted to enter the exact partition or queue name (e.g. batch, gpu, long).
- If N, no change is made and the pipeline uses the default partition settings in each slurms/*.slurm script.

Outcome:
- Exports PARTITION="–partition=your_partition " for use in all subsequent SBATCH commands.
- Provides a single-point configuration for cluster queue selection, avoiding manual edits in multiple SLURM scripts.
- if N selected, partition can be manually adjusted from within the slurm files.

Output

The output of running simulation_setup.sh is a directory named *Protein*_simulation where *Protein* is the name of the structure given as input to the script without the '.pdb' tag. This directory contains many subdirectories for each stage of the MD simulation. The file configurations.txt contains all the relevant environment variables needed to run the simulation. The file `monitor.txt` monitors when the simulation repeats have finished in order to initiate post-processing.

run_MD.sh

The `run_MD.sh` script executes the full GROMACS-based simulation workflow using the configurations generated by `simulation_setup.sh`. Invoke it with:

```
sh run_MD.sh <Protein_simulation_directory>
```

Upon launch, the script:

1. Sources the `configurations.sh` to import all environment variables
2. Generates GROMACS compatible structure and topology files via `gmx pdb2gmx`. Based on the variables stored in `configurations.txt`, you may be prompted here to define disulfide bonds and terminal cap selections.
3. Solvates the box and adds ions automatically, according to your chosen box geometry and ionic strength
4. Submits SLURM jobs for:
    - Energy minimisation
    - NVT equilibration
    - NPT equilibration
    - Three independent production runs
5. Manages dependencies so that each stage begins only after the previous one completes successfully
6. Monitors for checkpoint files and, if a job approaches its walltime limit, automatically resubmits continuation jobs
7. Upon completion of all three simulation repeats, initiates post-processing and analysis scripts

All compute-intensive steps are designed for dispatch through the relevant cluster's SLURM scheduler, with resource requests and email notifications configured as per the setup choices. Progress and GROMACS outputs are streamed to log files in each simulation subdirectory.

Simulation subdirectories

Each major step in the MD pipeline produces its own subfolder under the main `SIMULATION_DIR`. A brief overview:

`pdb2gmx/`

- Houses all GROMACS-compatible files generated by `gmx pdb2gmx`, including:
  - Structure (`*.gro`)
  - Topology (`*.top`)
  - Force-field includes (`*.itp`)
- These files define the initial coordinates, connectivity, and parameter assignments for the system

`solvate/`

- Contains the solvated system inside the chosen periodic box
- Note: the protein may not yet be centered; final centering occurs during conversions with `gmx trjconv` using the `.tpr` file

`genion/`

- Stores the neutralised and ion-adjusted system, with `gmx genion` adding Na^+/Cl^- to achieve the user-specified ionic strength

`minim1/`

- Performs the first energy minimisation using the steepest-descent algorithm
- Output structure: `minim1.gro`

minim2/

- Conducts a second minimisation using the conjugate-gradient method for finer refinement
- Output structure: `minim2.gro`

nvt/

- Equilibrates the system in the canonical (NVT) ensemble
- Key outputs: structure `nvt.gro` and trajectory `nvt.xtc`

npt/

- Equilibrates the system in the isothermal–isobaric (NPT) ensemble
- Key outputs: structure `npt.gro` and trajectory `npt.xtc`

production/ Contains all production-phase outputs and derived analyses:

- **rep1/, rep2/, rep3/**
  - Raw MD outputs: `md.xtc`, `md.gro`, `md.cpt`, etc.
  - `processed/`: stripped-solvent trajectories and topologies for downstream analysis. Key outputs: `final.pdb`, `final.xtc`, `nosolv.tpr`.
  - `standard_analysis/`: standard MD metrics (RMSD, radius of gyration, SASA, hydrogen-bond counts, total energy) plotted in `.svg` and `.pdf` formats from `.xvg` files
  - `extract_frames/`: PDB snapshots extracted from the processed trajectories at 1 ns intervals, used for synthetic SAXS computations with CRYSOL. χ^2 vs Rg plots computed for each repeat. For more information on CRYSOL, visit https://www.embl-hamburg.de/biosaxs/crysol.html.
  - If `SYSTEM=Protein-ligand`, `prot-lig_analysis` is generated containing ligand-centric analyses
- **combined_traj/**

- A single, concatenated trajectory (`combined_aligned.xtc`) spanning all three repeats, aligned to a reference structure (`final.pdb`).
- `clustering/`: Results from the heirarchical clustering algorithm, CLoNe. Contains four subdirectories for clustering calculations pefromed at varying *pdc* values. For more information on CLoNe, visit https://github.com/LBM-EPFL/CLoNe, https://doi.org/10.1093/bioinformatics/btaa742.
- `extract_frames/`: PDB snapshots at 1 ns intervals extracted from the combined trajectory for SAXS fitting.
  * SAXS-scored principal component analysis
  * Trajectories of the first three principal components, saved as c-alpha PDB files.
- `GA001/`: Ensemble reweighting outputs (GAJOE), including weighted-ensemble fits and R_g distribution comparisons. For more information on GAJOE visit https://www.embl-hamburg.de/biosaxs/manuals/atsas-2.7/eom.html

Examples

The directory `examples/` contains various examples runs demonstrating the full versatility of the pipeline:

- `SYSTEM=Protein` and `SYSTEM=Protein-ligand`
- With/without SAXS data
- Prepared with Protein Preparation Wizard, CHARMM-GUI, or H++

The protocols used to run each example are found within the relevant directory as `protocol.txt`.